

- Description

This module manages a smart electric vehicle (EV) charging fee calculation system, ensuring that:

1. The system continuously monitors the total energy consumption (in kWh) during a charging session.
2. A base fee is applied per kWh for energy consumption up to a predefined threshold.
3. If consumption exceeds the threshold, a higher fee (including a penalty rate) is applied to the excess energy.

This system helps ensure fair and efficient billing for EV charging, encouraging optimized energy usage.

- Specification

module SmartEVChargingFeeCalculation

var currentConsumption, baseRate, penaltyRate, threshold: real

process addConsumption(kWh: real) returns (totalFee: real, currentConsumption: real)

functional scenario form:

currentConsumption <= threshold && totalFee = currentConsumption * baseRate, Total fee calculated as: currentConsumption * baseRate.

||

currentConsumption > threshold && totalFee = threshold * baseRate + (currentConsumption - threshold) * (baseRate + penaltyRate), output "Total Consumption: " + currentConsumption + " kWh, Total Fee: \$" + totalFee, Total fee calculated as: threshold * baseRate + (currentConsumption - threshold) * (baseRate + penaltyRate).

end process

end module

- Java Code

```
package code;
class SmartEVCharger {
    private double baseRate;
    private double penaltyRate;
    private double threshold;
    private double currentConsumption;
    public SmartEVCharger(double baseRate, double penaltyRate, double threshold) {
        this.baseRate = baseRate;
        this.penaltyRate = penaltyRate;
        this.threshold = threshold;
        this.currentConsumption = 0.0;
    }
}
```

```

    public void addConsumption(double kWh) {
        if (kWh < 0) {
            System.out.println("Invalid consumption value. Must be positive.");
            return;
        }
        currentConsumption += kWh;
        double fee;
        if (currentConsumption <= threshold) {
            fee = currentConsumption * baseRate;
        } else {
            fee = threshold * baseRate + (currentConsumption - threshold) * (baseRate + penaltyRate);
        }
        System.out.printf("Current Consumption: %.2f kWh, Calculated Fee: $%.2f\n", currentConsumption, fee);
    }

    public double getCurrentConsumption() {
        return currentConsumption;
    }

    public double calculateFee() {
        if (currentConsumption <= threshold) {
            return currentConsumption * baseRate;
        } else {
            return threshold * baseRate + (currentConsumption - threshold) * (baseRate + penaltyRate);
        }
    }
}

public class c76_SmartEVChargingSystem {
    public static void main(String[] args) {
        SmartEVCharger charger = new SmartEVCharger(0.20, 0.10, 50.0);
        // testcase-VT:
        //     charger.addConsumption(30.0);
        //     charger.addConsumption(15.0);
        //     charger.calculateFee();
        //     charger.addConsumption(10.0);
        //     charger.addConsumption(-10.0);
        //     charger.calculateFee();

        // testcase-FT:
        //     charger.addConsumption(4.1593684296814365E307);
        //     charger.addConsumption(1.5778637915857737E293);
    }
}

```

```
//      charger.addConsumption(7.683982623208027E307);
//      charger.addConsumption(0.0);
//      charger.addConsumption(-1.7906708960542598E308);
//      charger.calculateFee();

// testcase-Z3:
//      charger.addConsumption(30.0);
//      charger.addConsumption(15.0);
//      charger.calculateFee();
//      charger.addConsumption(10.0);
//      charger.addConsumption(-10.0);
//      charger.calculateFee();

// testcase-UVT:
charger.addConsumption(30.0);
charger.addConsumption(15.0);
charger.calculateFee();
charger.addConsumption(10.0);
charger.addConsumption(-10.0);
charger.calculateFee();
    }
}
```

- Coverage report

1. Test case number: 6
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.7 %	6	836	842
✓ src	0.7 %	6	836	842
✓ code	0.7 %	6	836	842
✓ c76_SmartEVChargingSystem.java	100.0 %	6	0	6
> c76_SmartEVChargingSystem		0	0	0
> SmartEVCharger	100.0 %	6	0	6

3. Vibration testing instruction coverage: 95.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	0.6 %	134	20,898	21,032
✓ src	0.6 %	134	20,898	21,032
✓ code	0.6 %	134	20,898	21,032
✓ c76_SmartEVChargingSystem.java	95.7 %	134	6	140
> c76_SmartEVChargingSystem	89.7 %	26	3	29
> SmartEVCharger	97.3 %	108	3	111

4. Fuzz testing branch coverage: 66.7%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.5 %	4	838	842
✓ src	0.5 %	4	838	842
✓ code	0.5 %	4	838	842
✓ c76_SmartEVChargingSystem.java	66.7 %	4	2	6
> c76_SmartEVChargingSystem		0	0	0
> SmartEVCharger	66.7 %	4	2	6

5. Fuzz testing instruction coverage: 86.4%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.6 %	121	20,911	21,032
▼ src	0.6 %	121	20,911	21,032
▼ code	0.6 %	121	20,911	21,032
▼ c76_SmartEVChargingSystem.java	86.4 %	121	19	140
> c76_SmartEVChargingSystem	89.7 %	26	3	29
> SmartEVCharger	85.6 %	95	16	111

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.7 %	6	836	842
▼ src	0.7 %	6	836	842
▼ code	0.7 %	6	836	842
▼ c76_SmartEVChargingSystem.java	100.0 %	6	0	6
> c76_SmartEVChargingSystem		0	0	0
> SmartEVCharger	100.0 %	6	0	6

7. Z3 instruction coverage: 95.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.6 %	134	20,898	21,032
▼ src	0.6 %	134	20,898	21,032
▼ code	0.6 %	134	20,898	21,032
▼ c76_SmartEVChargingSystem.java	95.7 %	134	6	140
> c76_SmartEVChargingSystem	89.7 %	26	3	29
> SmartEVCharger	97.3 %	108	3	111

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.7 %	6	836	842
▼ src	0.7 %	6	836	842
▼ code	0.7 %	6	836	842
▼ c76_SmartEVChargingSystem.java	100.0 %	6	0	6
> c76_SmartEVChargingSystem		0	0	0
> SmartEVCharger	100.0 %	6	0	6

9. Unimproved V-method instruction coverage: 95.7%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.6 %	134	20,898	21,032
▼ src	0.6 %	134	20,898	21,032
▼ code	0.6 %	134	20,898	21,032
▼ c76_SmartEVChargingSystem.java	95.7 %	134	6	140
> c76_SmartEVChargingSystem	89.7 %	26	3	29
> SmartEVCharger	97.3 %	108	3	111